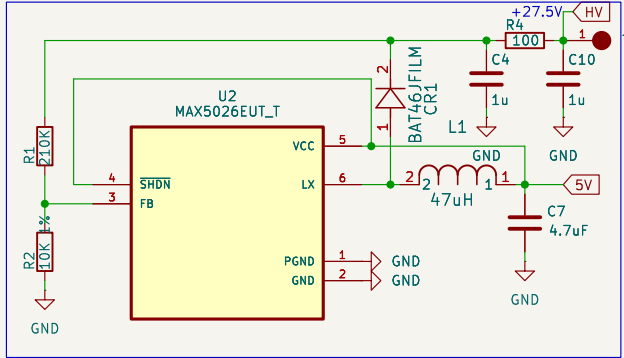


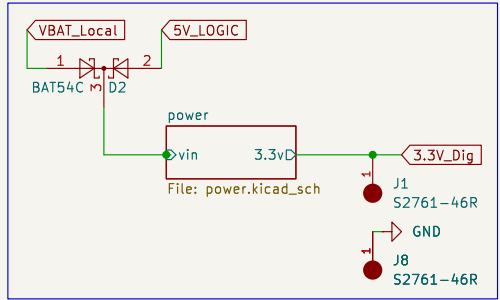
DC-DC Booster

Takes the VCC line, and increases the voltage to +32.5V.
This HV line is used to provide the reverse bias to the SiPM. Accounts for potential -1.29v vibas drift from 20C to -40



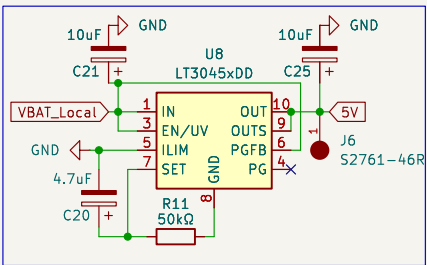
+3v3

Takes the battery line and regulates the voltage to 3.3v for the logic electronics



+5V

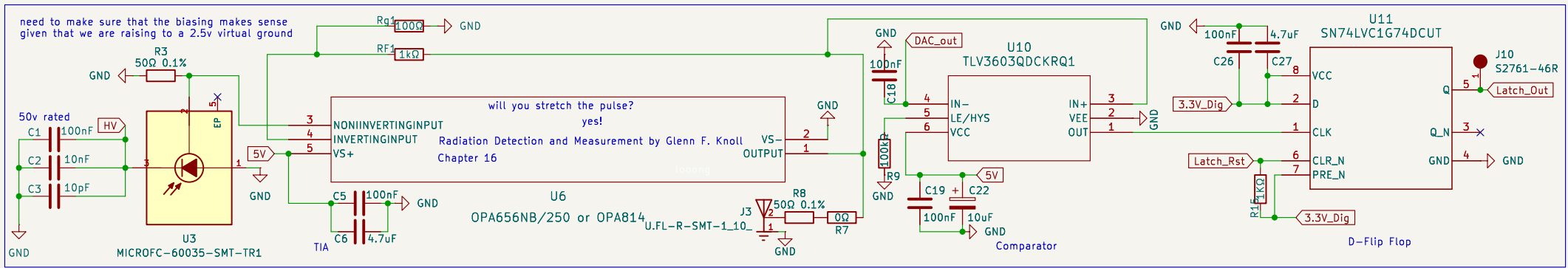
This creates the 5V supply that we use for the reference for the TIA



this is going up 9km, is the bias voltage ok?
yes.

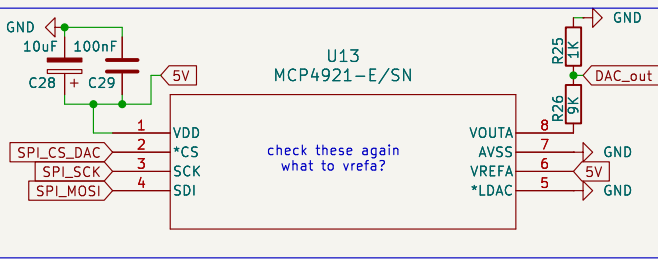
Readout

Amplifies, detects and registers the current pulse from the SiPM



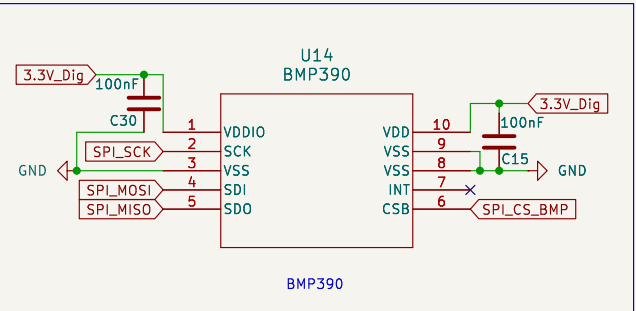
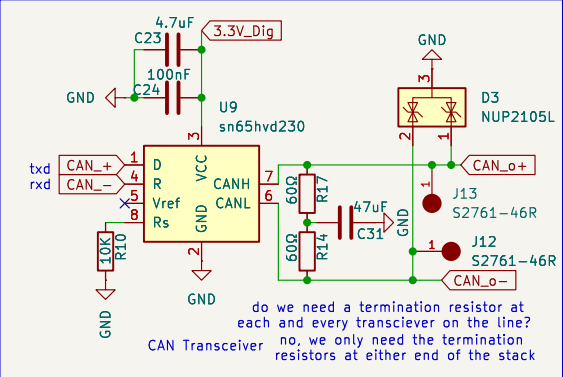
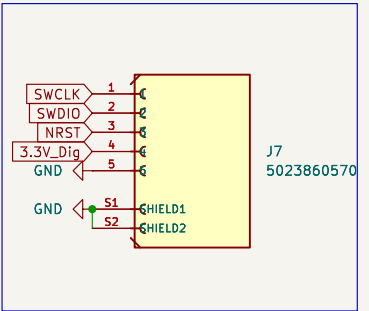
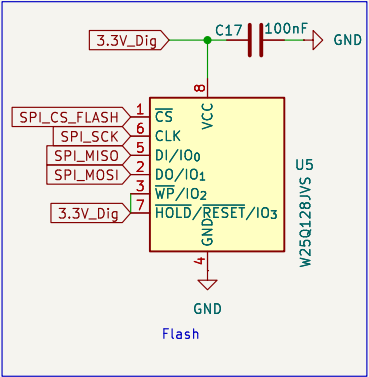
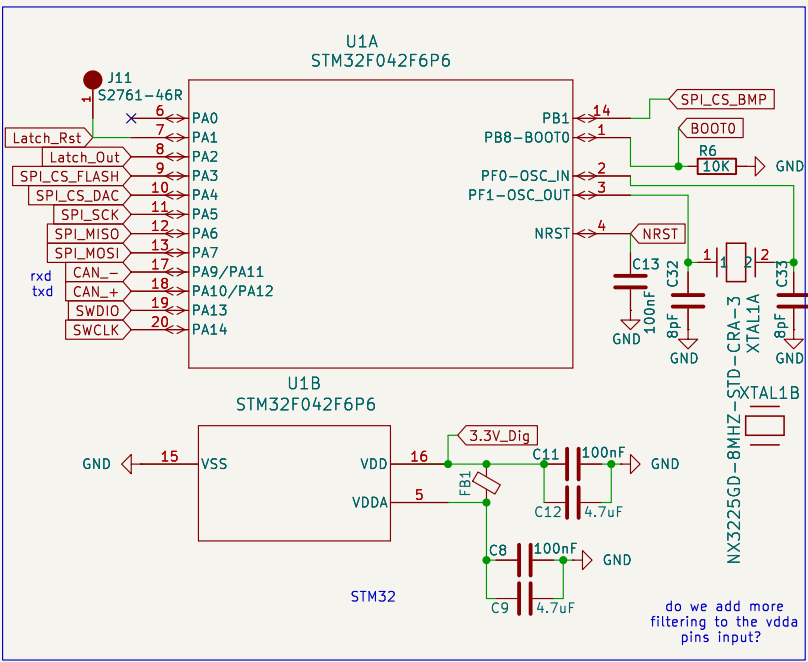
DAC

Sets a stable voltage reference for the comparator



Logging

STM32 mcu, nand flash, can transceiver, together operate the sensor, catch the pulses and log the data



Connectors

read it again <https://ie.rs-online.com/web/p/pcb-headers/2155821>

